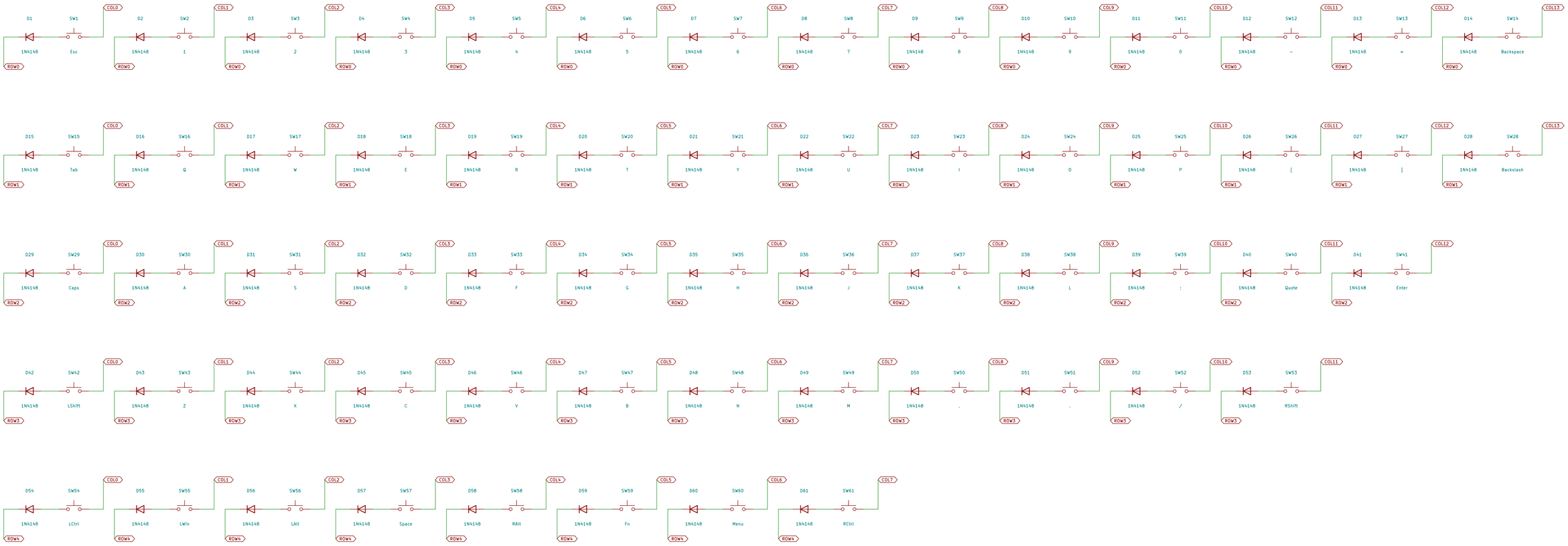
61-KEY ANSI / 60% KEYBOARD

5 x 14 matrix | 61 populated positions | COL2ROW | cathode stripe toward ROW

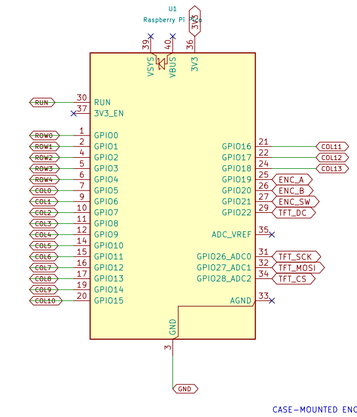


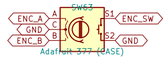


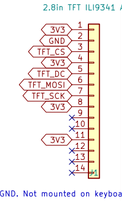
RESET: momentarily connects RUN to GND

BOOTSEL is on the Pico. Hold BOOTSEL while plugging in USB to flash.

CONTROLLER: ORIGINAL RP2040 PICO (NOT PICO W)







CASE-MOUNTED ENCODER: wire A/B/S1 to Pico pads 25/26/27; C/S2 to GND. Not mounted on keyboard PCB.

AA317 14-pin header: 1/4/8/11=3V3; 2=GND; 3=CS; 4=RESET; 5=DC; 6=MOSI; 7=SCK. Touch/SD disabled.
Backlight uses 3V3 supply (30–100mA module load), not a GPIO.
SD header J2 pin order: 1=SD_CS tied high; 2=MOSI, 3=MISO, 4=SCK left unconnected.



USB power + data through Pico onboard micro-USB. No external power input.

Unused Pico pins marked NC. GND pads 3/8/13/18/23/28/38 share GND.

Scan: ROW output low, COL inputs with pull-ups. Pressed key conducts COL -> switch -> diode -> ROW.

This sheet shows electrical matrix order, not physical key placement. See layout.csv for 19.05 mm pitch.

AI-assisted learning project

Sheet: /
File: keyboard61.kicad_sch

Title: 61-key ANSI keyboard — schematic only

Size: A2	Date: 2026-10-06	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1